Chapter 12 — Nuclear and Particle Physics

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — The nucleus and fundamental forces

Q.1 Describe the composition of the nucleus and compare the four fundamental forces.

The nucleus contains protons and neutrons, collectively called nucleons. The mass number A is the total number of nucleons, the atomic number Z is the number of protons, and the neutron number is N = A − Z. One atomic mass unit is equivalent to about 931 MeV. The nucleus is held together by the strong nuclear force, which overcomes the electrostatic repulsion between protons.

The four fundamental forces are the strong force, electromagnetic force, weak force and gravity. Their relative strengths are approximately 1, 10^{-2}, 10^{-13} and 10^{-38}. The strong force acts between quarks and holds nuclei together; the electromagnetic force acts between charges; the weak force is responsible for beta decay; and gravity acts between masses.

Force	Relative strength	Role
Strong	1	Holds quarks and nuclei together
Electromagnetic	10^{-2}	Acts between electric charges
Weak	10^{-13}	Responsible for beta decay
Gravity	10^{-38}	Acts between masses

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Important Conceptual Short Questions — Topic 1

1. Why is the strong force needed inside the nucleus?

Protons repel each other electrostatically. The strong nuclear force is much stronger at short range and binds protons and neutrons together, making the nucleus stable.

2. How are mass number and atomic number related to the nucleus?

The atomic number Z gives the number of protons, and the mass number A gives the total number of protons and neutrons. The neutron number is therefore N = A − Z.

3. Why is gravity negligible inside the nucleus?

The masses of nucleons are extremely small, and gravity is about 10^{-38} times weaker than the strong force. Its effect inside the nucleus is therefore negligible.

Topic 2 — Matter, antimatter and radioactivity

Q.2 Explain matter and antimatter, pair production and annihilation, and compare alpha, beta and gamma radiations.

Every particle has an antiparticle with the same mass but opposite charge, for example the positron is the antiparticle of the electron. In pair production, a gamma photon of at least 1.02 MeV produces an electron–positron pair near a nucleus. In annihilation, an electron and positron destroy each other and produce two gamma photons of 0.51 MeV each.

In alpha decay the nucleus emits a helium nucleus, so A decreases by 4 and Z by 2. In beta-minus decay a neutron changes into a proton, so Z increases by 1 and A is unchanged; an electron and an antineutrino are emitted. In beta-plus decay a proton changes into a neutron, emitting a positron and a neutrino. Gamma radiation is emitted when an excited nucleus loses energy without changing A or Z. Alpha particles have discrete energies, while beta particles have a continuous range because the energy is shared with a neutrino or antineutrino.

Textbook check (p. 260): alpha decay is linked to a nucleus having more protons than neutrons. Alpha emitters usually have more neutrons than protons; the cause is the large size of the nucleus. Also p. 262 lists gamma energy as variable, but gamma rays from a given nucleus have discrete energies.

Important Conceptual Short Questions — Topic 2

1. Why do beta particles have a continuous energy spectrum?

The decay energy is shared between the beta particle and a neutrino or antineutrino in varying proportions. The beta particle can therefore have any energy from zero up to a maximum.

2. Why are beta particles not orbital electrons?

Beta particles are created inside the nucleus at the moment of decay when a neutron changes into a proton or a proton changes into a neutron. They do not come from the electron shells.

3. Why is a nucleus needed for pair production?

The nucleus absorbs some momentum so that both energy and momentum can be conserved. A single photon cannot produce a pair in empty space while conserving momentum.

Topic 3 — Fundamental particles and quarks

Q.3 Classify fundamental particles and describe the quark model of hadrons.

Fundamental particles have no internal structure. They include six leptons, such as the electron and neutrinos, and six quarks: up, down, strange, charm, top and bottom. Hadrons are particles made of quarks and feel the strong force. Baryons, such as protons and neutrons, consist of three quarks; mesons consist of a quark and an antiquark. The proton is uud and the neutron is udd.

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Quarks carry colour charge, and the strong force between them is described by quantum chromodynamics. In beta-minus decay a down quark changes into an up quark: $d \rightarrow u + e^{-} + \bar{\nu}$, converting a neutron into a proton. Exchange particles or gauge bosons carry the forces: photons for electromagnetism, gluons for the strong force, W and Z bosons for the weak force, and the graviton is proposed for gravity.

Important Conceptual Short Questions — Topic 3

1. Why are protons and neutrons not fundamental particles?

Each is made of three quarks, so it has internal structure. Fundamental particles such as electrons and quarks are not made of smaller constituents.

2. How does the quark model explain beta-minus decay?

A down quark in the neutron changes into an up quark, turning udd into uud. The neutron becomes a proton and an electron and an antineutrino are emitted.

3. Why are quarks never found in isolation?

The strong force between quarks does not weaken with distance in the usual way. Trying to separate them requires so much energy that new quark–antiquark pairs are produced, so only colour-neutral hadrons are observed.

Topic 4 — Higgs boson, conservation laws and the Standard Model

Q.4 Explain the Higgs boson, conservation laws in particle reactions and the limits of the Standard Model.

The Higgs boson is associated with the Higgs field, which gives mass to fundamental particles that interact with it. It was discovered at CERN in 2012 with a mass of about 125 GeV/c^2. Particles that interact strongly with the field, such as W and Z bosons, have large mass, while photons do not interact and have zero rest mass.

In particle reactions, charge, nucleon number, energy and momentum are conserved. The Standard Model successfully describes leptons, quarks and the electromagnetic, weak and strong forces, but it does not include gravity and does not explain dark matter or the matter–antimatter asymmetry of the universe. Most of the observable matter in the universe is plasma, and theories such as field theory and string theory attempt to go beyond the Standard Model.

Textbook check (p. 266): the free neutron half-life is given as 900 s. It is about 610 s; 880 s is the average lifetime. Also p. 272 uses electron mass 0.000555 u, giving 0.517 MeV; the accepted value about 0.000549 u gives 0.511 MeV.

Important Conceptual Short Questions — Topic 4

1. Why is the Higgs boson important in the Standard Model?

It confirms the existence of the Higgs field, which explains how fundamental particles acquire mass. Without this mechanism the Standard Model could not account for the masses of W and Z bosons and other particles.

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2. Why does the universe contain more matter than antimatter?

The early universe produced almost equal amounts, but a small imbalance favoured matter. The surviving matter forms the universe we see; the exact cause of the asymmetry is not yet fully explained.

3. Why is the Standard Model not a complete theory of everything?

It does not include gravity, dark matter or the matter–antimatter asymmetry. It also leaves open questions about how mass and force arise at a deeper level, so further theories are being explored.

4. How are charge and nucleon number conserved in nuclear reactions?

The sum of the atomic numbers before the reaction equals the sum after it, so charge is conserved. The sum of the mass numbers also remains the same, so nucleon number is conserved. These checks are used to identify unknown particles in a reaction.

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